Time Series Analysis

Chapter 02
Fundamental Mathematical Concepts

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2 Means, Variances, and Covariances

3 Stationarity

Time Series and Stochastic Processes

- An observed time series can be modeled with a *stochastic process*: a sequence of random variables taken across time $\{Y_t, t = ..., -2, -1, 0, 1, 2, ...\}$.
- The probability structure of a stochastic process is determined by the set of all joint distributions of all finite set of these r.v.'s.
- If these joint distribution are *multivariate normal*, it is simpler: knowing the means, variances, and covariances of the Y_t's tells us everything about the joint distributions.

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Mean, Variances and Covariances

- The **mean function** of a stochastic process $\{Y_t\}$

$$\mu_t = E(Y_t) \text{ for all } t$$

  gives the expected value of the process at time t
- μ_t could be vary across time.
- The **autocovariance function** is denoted:

$$\gamma_{t,s} = cov(Y_t, Y_s) = E[(Y_t - \mu_t)(Y_s - \mu_s)] = E(Y_t Y_s) - \mu_t \mu_s \text{ for } t, s$$

- The **autocorrelation function** is denoted:

$$\rho_{t,s} = corr(Y_t, Y_s) = \frac{cov(Y_t, Y_s)}{\sqrt{var(Y_t) var(Y_s)}} \text{ for } t, s$$

Interpreting Correlation and Covariances

- Both autocovariance and autocorrelation measure the linear independence between the process's values at two different times.
- The autocorrelation is scaled to be between -1 and 1 and is easier to interpret.
- Many time series processes have positive autocovariance and correlation:
  - if $\gamma_{t,s} > 0$, then: If Y_t is large (small), then Y_s tends to be large (small).
  - if $\gamma_{t,s} < 0$, then: If Y_t is large, then Y_s tends to be small (and vice versa).
- Note $\gamma_{t,t} = var(Y_t) = E[(Y_t - \mu_t)^2] = E(Y_t^2) - \mu_t^2$.

Important Results

- If $c_1, c_2, ..., c_m$ and $d_1, d_2, ..., d_n$ are constants and $t_1, t_2, ..., t_m$ and $s_1, s_2, ..., s_n$ are time points

$$cov\left[\sum_{i=1}^{m} c_i Y_{t_i}, \sum_{j=1}^{n} d_j Y_{s_j}\right] = \sum_{i=1}^{m}\sum_{j=1}^{n} c_i d_j cov(Y_{t_i}, Y_{s_j})$$

A special case of this result:

$$var\left[\sum_{i=1}^{n} c_i Y_{t_i}\right] = \sum_{i=1}^{n} c_i^2 var(Y_{t_i}) + 2\sum_{i=2}^{n}\sum_{j=1}^{i-1} c_i c_j cov(Y_{t_i}, Y_{t_j})$$

Example: Calculate a Correlation of Linear Combination of r.v.'s

Example: Let Y_1 be a r.v. with $E(Y_1) = 0$ and $var(Y_1) = 4$ and Y_2 be a r.v. with $E(Y_2) = 0$ and $var(Y_2) = 9$, and let $cov(Y_1, Y_2) = 1$. Find $corr(2Y_1 + Y_2, 3Y_1 - 2Y_2)$.

Ans: $corr(2Y_1 + Y_2, 3Y_1 - 2Y_2) \approx 0.12$

A simple time series process: The Random Walk

- Let $e_1, e_2, ...$ be a sequence of independent and identically distributed (iid) r.v.'s, each having mean 0 and variance σ_e^2.
- Consider the time series:

$$\begin{aligned} Y_1 &= e_1 \\ Y_2 &= e_1 + e_2 \\ &\vdots \\ Y_t &= e_1 + e_2 + ... + e_t \end{aligned}$$

- In other words, $Y_t = Y_{t-1} + e_t$, where initially $Y_1 = e_1$.
- Then Y_t is the position on a number line (at time t) of a walker who is taking random (forward or backward) steps (of sizes $e_1, e_2, ..., e_t$) along the number line.

Properties of the Random Walk

- For this random walk process, the mean function $\mu_t = 0$ for all t, since

$$E(Y_t) = E(e_1 + e_2 + ... + e_t) = 0 + 0 + ... + 0 = 0$$

and

$$var(Y_t) = var(e_1 + e_2 + ... + e_t) = \sigma_e^2 + \sigma_e^2 + ... + \sigma_e^2 = t\sigma_e^2$$

since all the e_i's are independent.

Autocovariance Function of the Random Walk

- For $1 \leq t \leq s$,

$$\begin{aligned}\gamma_{t,s} &= cov(Y_t, Y_s) \\ &= cov(e_1 + e_2 + ... + e_t, e_1 + e_2 + ... + e_t + e_{t+1} + ... + e_s)\end{aligned}$$

- From the formula for covariance of sums of r.v.'s, we have

$$cov(Y_t, Y_s) = \sum_{i=1}^{s} \sum_{j=1}^{t} cov(e_i, e_j),$$

but these covariance terms are zero except when $i = j$, so $cov(Y_t, Y_s) = t\sigma_e^2$, for $1 \leq t \leq s$.

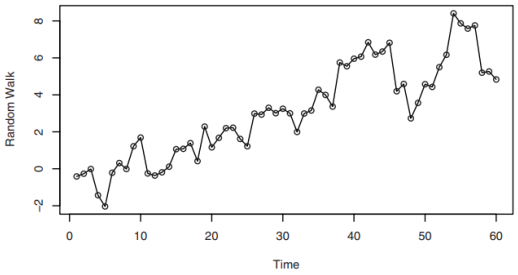
Autocorrelation Function of the Random Walk

- Autocorrelation function is easily found to be $\sqrt{t/s}$, for $1 \leq t \leq s$.
- From this, note $corr(Y_1, Y_2) = \sqrt{1/2} = 0.707$;

  $corr(Y_{24}, Y_{25}) = \sqrt{25/25} = 0.98$;

  $corr(Y_1, Y_{25}) = \sqrt{1/25} = 0.20$.
- Values close in time are more strongly correlated than values far apart in time.
- And neighboring values later in the process are more strongly correlated than neighboring values early in the process.

Exhibit 2.1: Time Series Plot of a Random Walk



- A simulated random walk where e's were selected from a standard normal distribution.
- Even the mean function is zero for all time points, the variance increase over time and the correlation between process values near by in time is nearly 1.
- We should expect long excursions of the process away from the mean level of zero.

R-code

- The simple random walk process provides a good model (at least to a first approximation) for phenomena as diverse as the movement of common stock price, and the position of small particles suspended in a fluid so-called Brownian motion.

```
win.graph(width=4.875, height=2.5,pointsize=8)
data(rwalk) # rwalk contains a simulated random walk
plot(rwalk,type='o',ylab='Random Walk')
```

A Simple Moving Average Process

- Let $e_1, e_2, \ldots$ be a sequence of independent and identically distributed (iid) r.v.'s, each having mean 0 and variance σ_e^2.
- Consider the time series:
$$Y_t = \frac{e_t + e_{t-1}}{2}.$$
- For this moving average process, the mean function $\mu_t = 0$ for all t, since $E(Y_t) = E[(e_t + e_{t-1})/2] = 0.5E[e_t + e_{t-1}] = 0$, and $var(Y_t) = var[(e_t + e_{t-1})/2] = 0.25var[e_t + e_{t-1}] = 0.25 \times 2\sigma_e^2 = 0.5\sigma_e^2$ since all the e_i's are independent.

Autocovariance Function of the Moving Average

$$\begin{aligned} cov(Y_t, Y_{t-1}) &= cov\left(\frac{e_t + e_{t-1}}{2}, \frac{e_{t-1} + e_{t-2}}{2}\right) \\ &= 0.25[cov(e_t, e_{t-1}) + cov(e_t, e_{t-2}) + cov(e_{t-1}, e_{t-1}) + cov(e_{t-1}, e_{t-2})] \\ &= 0.25[0 + 0 + cov(e_{t-1}, e_{t-1}) + 0] = 0.25\sigma_e^2 \end{aligned}$$

- And
$$cov(Y_t, Y_{t-2}) = cov\left(\frac{e_t + e_{t-1}}{2}, \frac{e_{t-2} + e_{t-3}}{2}\right) = 0$$
since there are no overlapping e terms here, and all the e_i's are independent.
- Similarly, $cov(Y_t, Y_{t-k}) = 0$ for all $k > 1$.

Autocorrelation Function of the Moving Average

- From this, note

$$\rho_{t,s} \begin{cases} 1, \text{ for } |t-s| = 0 \\ 0.5, \text{ for } |t-s| = 1 \\ 0, \text{ for } |t-s| > 1 \end{cases}$$

- Since $0.25\sigma_e^2/0.5\sigma_e^2 = 0.5$. Notice that $\rho_{2,1} = \rho_{3,2} = \rho_{4,3} = \rho_{9.8} = 0.5$.
- So $\rho_{t,t-1}$ is the same no matter what t is, and in fact, for any k, $p_{t,t-k}$ is the same no matter what t is.
- This is related to the concept of *stationarity*.

Exercise

2.1) Suppose $E(X) = 2, Var(X) = 9, E(Y) = 0, Var(Y) = 4$, and $Corr(X, Y) = 0.25$. Find:

a) $Var(X + Y)$

b) $Cov(X, X + Y)$

c) $Corr(X + Y, X - Y)$

2.2) If X and Y are dependent but $Var(X) = Var(Y)$, find $Cov(X + Y, X - Y)$.

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Stationarity

- If a process is stationary, this implies that the laws that govern the process do not change as time goes on.
- A process is *strictly stationary* if the entire joint distribution of n values is the same as the joint distribution of any other n time-shifted values of the process, no matter when the two sequences start.
- For example, with a stationary process, the joint distribution of Y_1, Y_3, Y_4 would be the same as the joint distribution of Y_6, Y_8, Y_9, and similarly for any such pairs of sequences.

Properties of Stationary Process

- Note that with a stationary process: $E(Y_t) = E(Y_{t-k})$ for all t and k, so this implies that the mean function of any stationary process is *constant* over time.
- Also, with a stationary process: $var(Y_t) = var(Y_{t-k})$ for all t and k, so the variance function of any stationary process is *constant* over time.
- Note: A function that is constant over time is one that does not depend on t.

More on Stationarity

- Also, if the process is stationary, the bivariate distribution of (Y_t, Y_s) is the same as the bivariate distribution of (Y_{t-k}, Y_{s-k}) for all t, s, k.
- So $cov(Y_t, Y_S) = cov(Y_{t-k}, Y_{s-k})$ for all t, s, k.
- Letting $k = s$, we have $cov(Y_t, Y_s) = cov(Y_{t-s}, Y_0)$; letting $k = t$, we have $cov(Y_t, Y_s) = cov(Y_0, Y_{s-t})$.
- The covariance between Y_t and Y_s depends on time only through the time difference $|t - s|$ and not otherwise on the actual times t and s.
- So $cov(Y_t, Y_s) = cov(Y_0, Y_{|t-s|})$.
- So for the stationary process, we can express our autocovariance and autocorrelation functions simply in terms of the time lag k:

$$\gamma_k = cov(Y_t, Y_{t-k}) \text{ and } \rho_k = corr(Y_t, Y_{t-k}).$$

Weak Stationarity

- A process is *weakly stationary* or *second-order stationary* if
  1. The mean function is constant over time, and
  2. $\gamma_{t,t-k} = \gamma_{0,k}$ for every time t and lag k.
- Any process that is strictly stationary is also weakly stationary.
- But a process could be weakly stationary and NOT strictly stationary.
- In the special case that all joint distributions for all the process are multivariate normal, then being weakly stationary is *equivalent* to being strictly stationary.

White Noise

- The *white noise process* is a simple example of a stationary process.
- Whit noise is simply a sequence of iid r.v.'s $\{e_t\}$.
- White noise is strictly stationary since

$$\begin{aligned} P[e_{t_1} \leq x_1, ..., e_{t_n} \leq x_n] &= P[e_{t_1} \leq x_1]...P[e_{t_n} \leq x_n] \\ &= P[e_{t_1-k} \leq x_1]...P[e_{t_n-k} \leq x_n] \\ &= P[e_{t_1-k} \leq x_1, ..., e_{t_n-k} \leq x_n] \end{aligned}$$

- Clearly, $\mu_t = E(e_t)$ is constant, and $\gamma_k = var(e_t) = \sigma_e^2$ for $k = 0$ and zero for any $k \neq 0$.

Examples of Stationary and Nonstationary Process

- The moving average process is another stationary process.
- The random walk process is not stationary. How can we see that?
- Its variance function is NOT constant, and its covariance function does NOT only depend on the time lag.
- What if we considered the *differences of successive* Y-values: $\nabla Y_t = Y_t - Y_{t-1}$?
- Since for the random walk, $\nabla Y_t = e_t$, or simply white noise, we see the *differenced* time series is stationary.
- This is common in practice: We can often transform nonstationary processes into stationary processes by *differencing*.

Exercise

2.3) Let X and a distribution with mean μ and variance σ^2, and let $Y_t = X$ for all t.

- Show that $\{Y_t\}$ is strictly and weakly stationary.
- Find the autocovariance function for $\{Y_t\}$.
- Sketch a "typical" time plot of Y_t.